WO

WO — Water Optimization

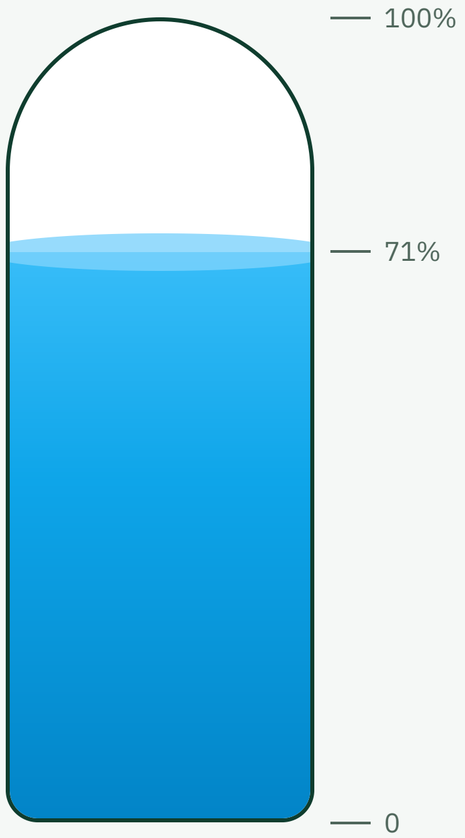
Smart Water Management for Agriculture: a knowledge-based decision support system for irrigation, hydroponics and land planning.



Introduction to Artificial Intelligence — course project

Reference text: Russell & Norvig, Artificial Intelligence: A Modern Approach

Agriculture accounts for most of the world's freshwater withdrawals



71%

Agriculture accounted for 71 per cent of total freshwater withdrawals globally in 2022.

23%

of global cultivated land was equipped for irrigation (2022), up from 21.5 per cent in 2015.

5,326 m³

global renewable freshwater resources per capita, a 7 per cent decline since 2015.

18%

global water stress in 2022; 733 million+ people live in countries with high or critical water stress.

0.69 USD/m³

agriculture's water-use efficiency (least efficient sector), improved 38 per cent 2015–2022.

Source for all figures: FAO AQUASTAT Water Data Snapshot 2025 — FAO newsroom, fao.org/newsroom/detail/renewable-water-availability-per-person-plunges-7-percent-in-a-decade-as-global-scarcity-deepens--fao-data-shows/en

WO turns a farm's own data into explained water decisions

Your data

Plantings, records, weather

Crops and planting dates, water use, hydroponic readings, a validated forecast.

Perception

Numbers become facts

FAO-56 requirement, then percepts such as `water_limited`: true, false or unknown.

Knowledge base

39 rules, sourced values

Horn rules, axioms and parameters stored in the database and editable by an admin.

Inference

Forward and backward chaining

Events run forward chaining; questions run backward chaining and find missing data.

Decision

Recommendation + why

Facts used, rules fired, why not the alternatives, what could change it.

No ML, no generative AI

Deterministic logic from the course textbook; the chatbot is pattern-based.

No invented data

The workspace starts empty; only the knowledge base is pre-loaded, with sources.

Honest about gaps

Missing input means "Insufficient data", with the list of what is needed.

Six objectives, each one implemented and testable

Calculate the requirement

FAO-56 single crop coefficient per planting, with every step stored and shown.

Recommend and explain

A knowledge-based agent decides; every decision cites facts, rules and alternatives.

Treat hydroponics as core

Measured water balance and per-system pH/EC ranges, never soil equations.

Keep working offline

Saved weather snapshots or manual ET0; "unavailable" instead of guessed values.

Plan with search and CSP

A* land distribution with an admissible heuristic; irrigation slots by backtracking.

Report with evidence

Calculated, actual, apparent and measured water kept apart; no assumed savings.

One platform, one agent at the centre

82

REST API operations (FastAPI, OpenAPI)

26

PostgreSQL tables, Alembic migrations

22

React pages in Arabic and English (RTL)

5

roles with backend-enforced permissions

Knowledge-based agent
Perception, rules, forward/backward chaining, explanations, KB validation, Logic Lab

Farms and plantings
32 FAO-56 reference crops

Water records and analytics
Totals, trends, anomalies, impact

Smart irrigation
Explained recommendation, CSP schedule

Hydroponics
Water balance, pH/EC, rules

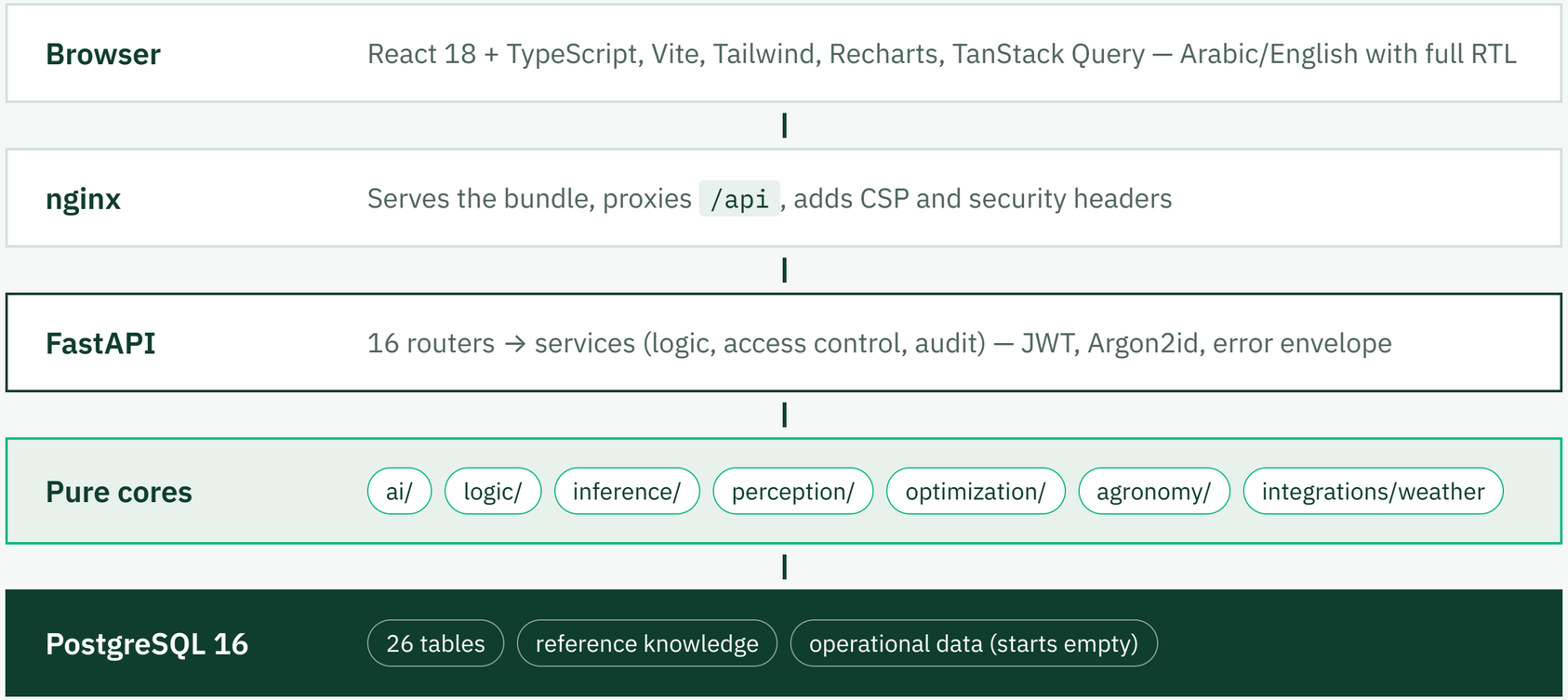
Weather
Open-Meteo with offline fallback

Assistant
Deterministic Arabic/English chatbot

Land distribution
A* with BFS/UCS/Greedy comparison

Alerts and recommendations
Rule-based, deduplicated, with evidence

Thin routers, testable cores



Routers

Parse the request, call one service, return a schema.

Services

Load data, enforce roles and isolation, persist results and audit entries.

Cores

Plain functions: no database, no network — easy to test and demonstrate.

Open-Meteo

Optional forecast source. The only outbound call.

Soil irrigation follows FAO-56, step by step

Worked example computed by WO's code with illustrative inputs: tomato, 44 days after planting, 500 m², drip, ET0 6.2 mm, rain 2 mm, RHmin 30 %, wind 3 m/s.

1

Growth stage

`Table 11`

Development

day 14 of 40

2

Crop coefficient

`Eq. 66 + Eq. 62`

0.8076

Kc_mid adjusted 1.15 → 1.193

3

Crop ET

`ETc = Kc × ET0`

5.007 mm

0.8076 × 6.2

4

Effective rain

`P_eff = 0.8 × P`

1.600 mm

0.8 × 2.0

5

Net requirement

`NIR = max(0, ETc − P_eff)`

3.407 mm

× 500 m² = 1,703.45 L

6

Gross volume

`V = net / efficiency`

1,892.72 L

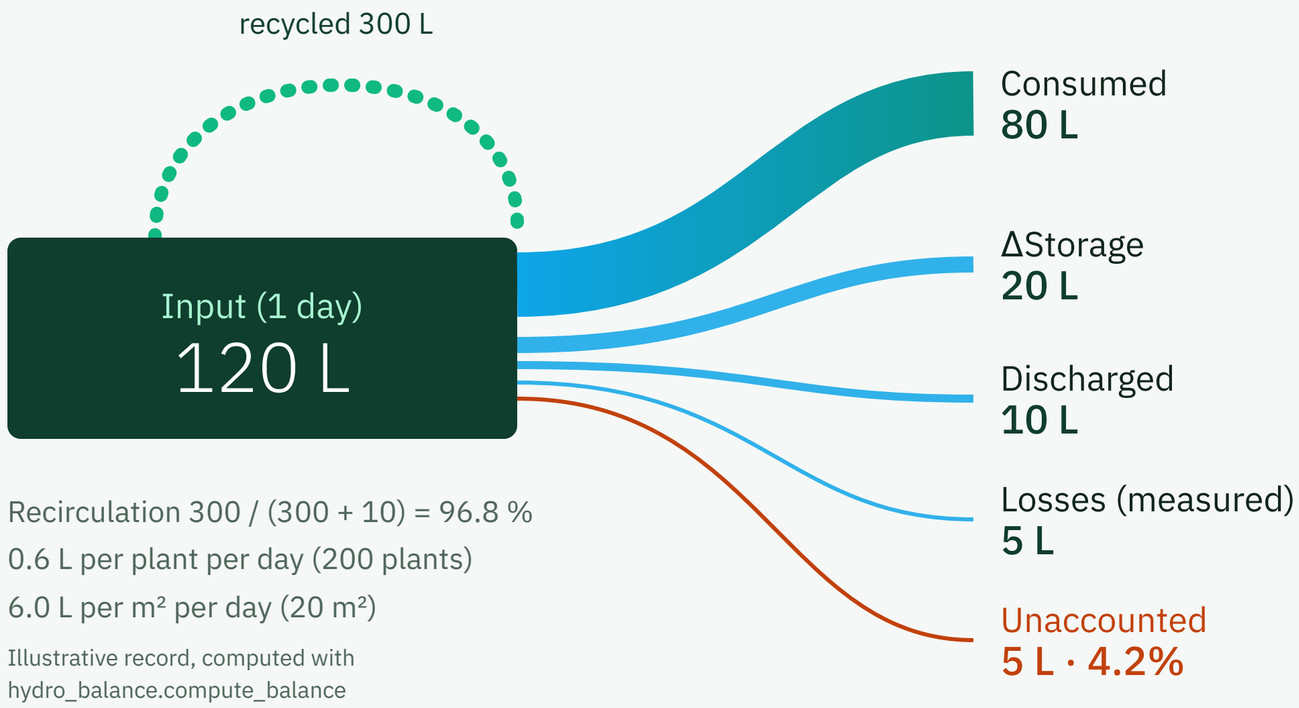
drip efficiency 0.90



Default application efficiencies from FAO Irrigation Water Management Training Manual No. 4 (methodology, not WO results). Method: FAO-56, Allen et al. (1998), Tables 11–12; every coefficient in the knowledge base carries its source.

Hydroponics is measured, not estimated

Input = Consumed + Discharged + ΔStorage + Losses + **Unaccounted**



Recirculation 300 / (300 + 10) = 96.8 %

0.6 L per plant per day (200 plants)

6.0 L per m² per day (20 m²)

Illustrative record, computed with hydro_balance.compute_balance

Missing terms are reported

Unaccounted water is computed only when consumed, discharged and both reservoir levels are recorded.

Ranges are configured per system

pH 7.2 above a configured 6.5 fires `R-HYD-002` and advice `R-HYD-010`. No range configured: no alert, "range not configured".

Reference profile with source

Lettuce: pH 5.6–6.0, EC 1.15–1.25 mS/cm (Cornell CEA Hydroponic Lettuce Handbook).

8 system types

NFT · DWC · Ebb & flow · Drip · Wick · Aeroponics · Raft · Substrate

Weather helps, but it is never required

Fresh

Fetch, validate, save

Open-Meteo daily ET0, rain, temperature, wind; ranges and array lengths checked; extended set retried as core on HTTP 400.

Cached

Latest valid snapshot

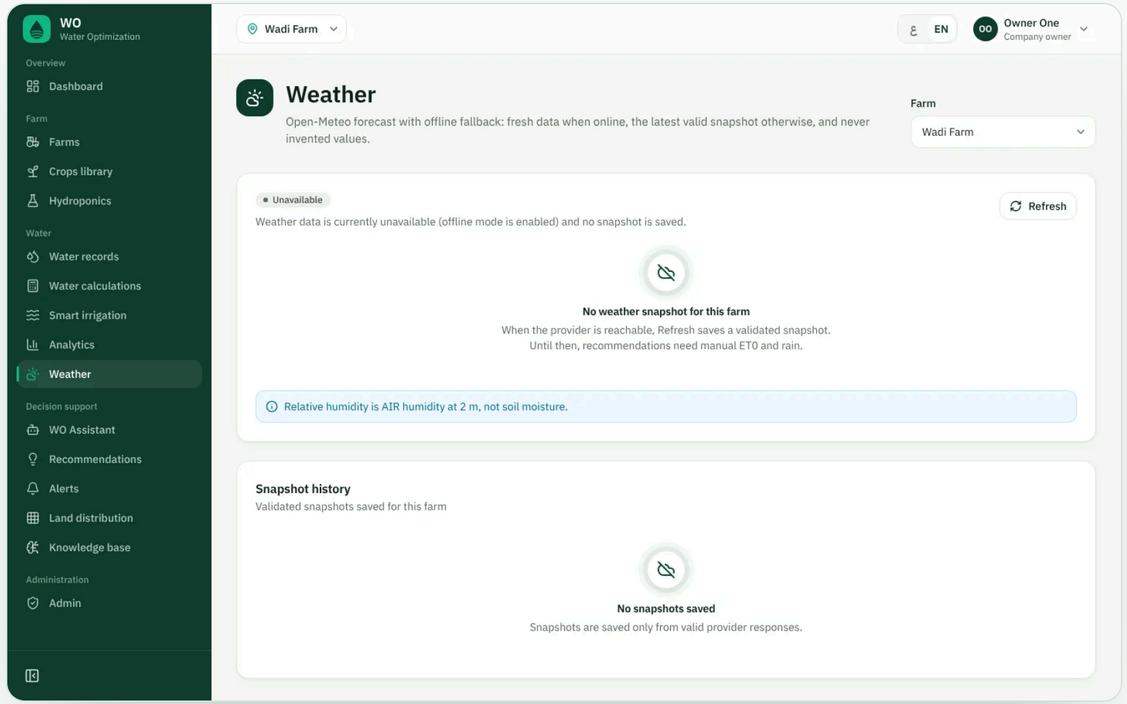
Used when offline; flagged stale after 12 h (a policy value in the knowledge base).

None

Unavailable, never invented

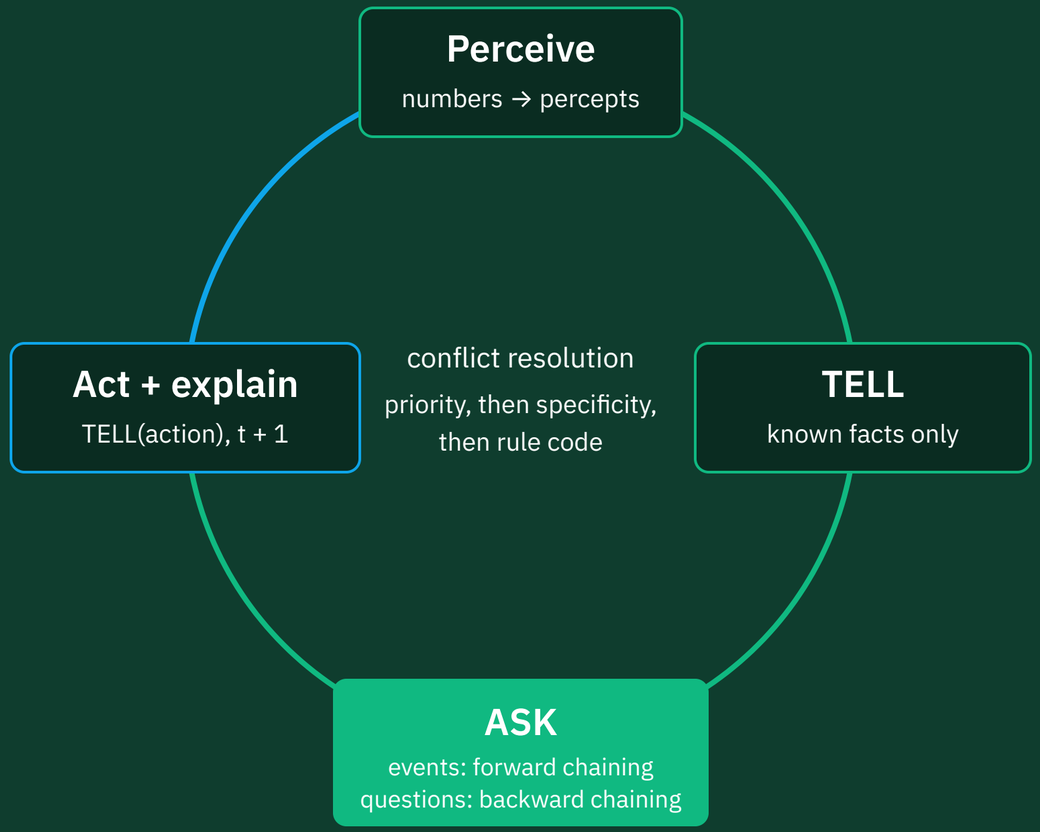
The agent reports ET0 and rain as missing; the user can enter them manually (data quality: manual ET0).

> Relative humidity is air humidity at 2 m. WO never treats it as soil moisture.



The real weather page with offline mode on and no snapshot: it says so, and asks for manual ET0 and rain.

A knowledge-based agent: TELL, then ASK



39
Horn rules

27
percepts

16
parameters

8
axioms

Unknown is not false

`~x` holds only when x is known false, so rules stay Horn clauses and chaining stays sound and complete.

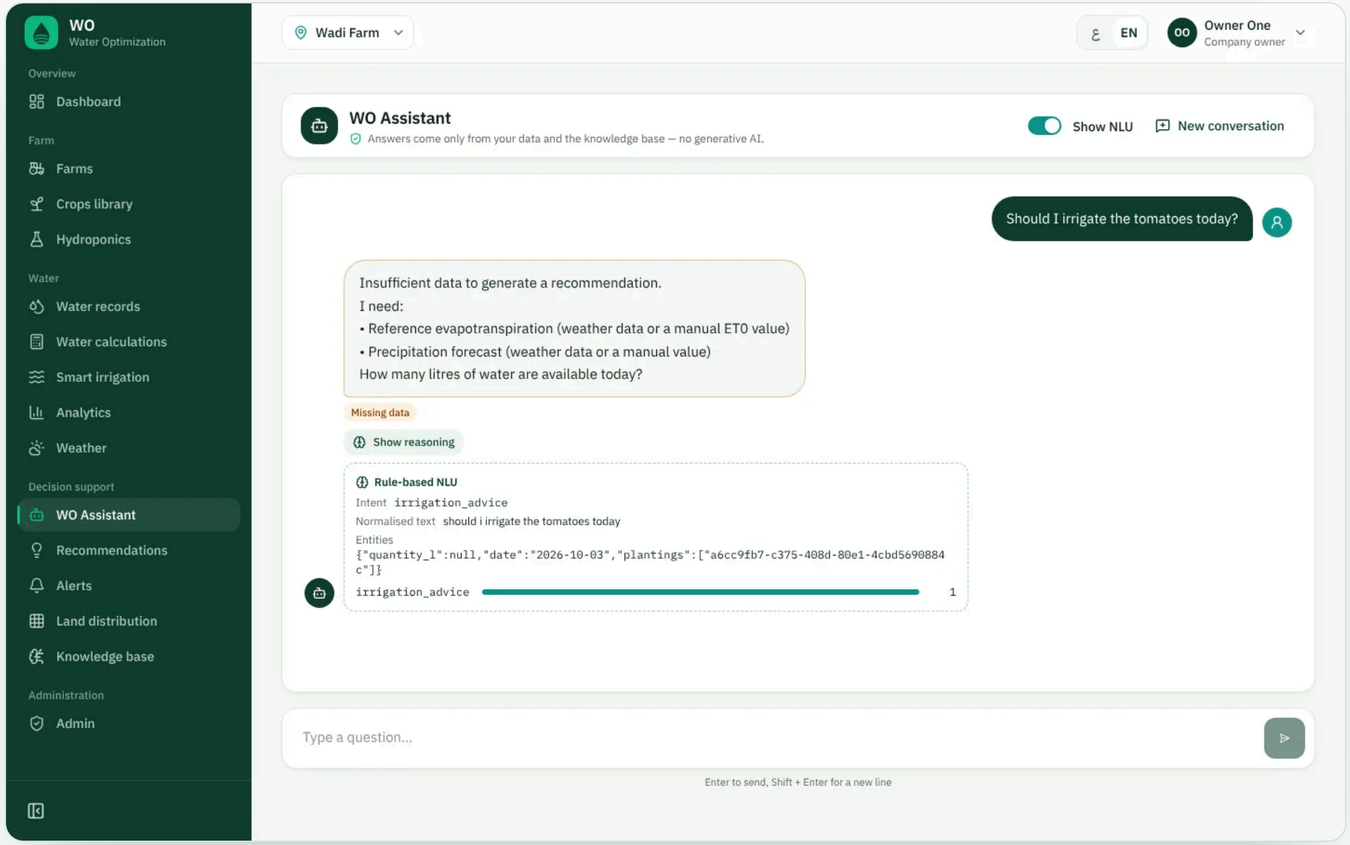
Every rule change is model-checked

Clark completion + DPLL: consistency, integrity, dead rules, conflicts, coverage. Shipped KB: 60 SAT checks, no conflict.

Logic Lab: five engines agree

Forward and backward chaining, TT-ENTAILS, DPLL and resolution on the same query.

An assistant that answers only from your data



Real screen: weather was offline, so the agent answered "Insufficient data" and asked for ET0, rain and available water.

1 Normalise
Strip tashkeel and tatweel, unify alef, taa marbuta, yaa; Arabic digits to 0–9.

2 Detect intent
10 intents, weighted Arabic and English patterns, threshold and tie rules.

3 Extract entities
Your crops and aliases, dates such as اليوم and بكرة, litres and m³.

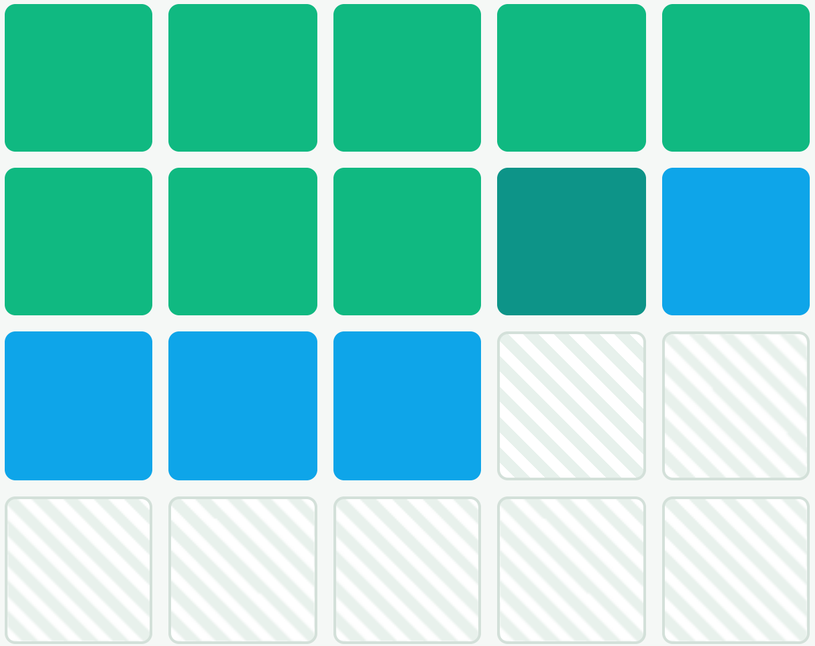
4 Fill slots from missing data
Two tomato plantings: "Which planting?" Weather offline: "Send ET0 and rain." No water figure: "How many litres?"

5 Run the agent, fill a template
Same rules and explanation as the irrigation page. No LLM.

هل أسقي الطماطم اليوم؟

normalised: هل اسقي الطماطم اليوم
`intent = irrigation_advice (3.5), date = today`

Land distribution as search: A* finds the optimum with 7 expansions



Tomato 8 Onion 1 Watermelon 4 Fallow 7

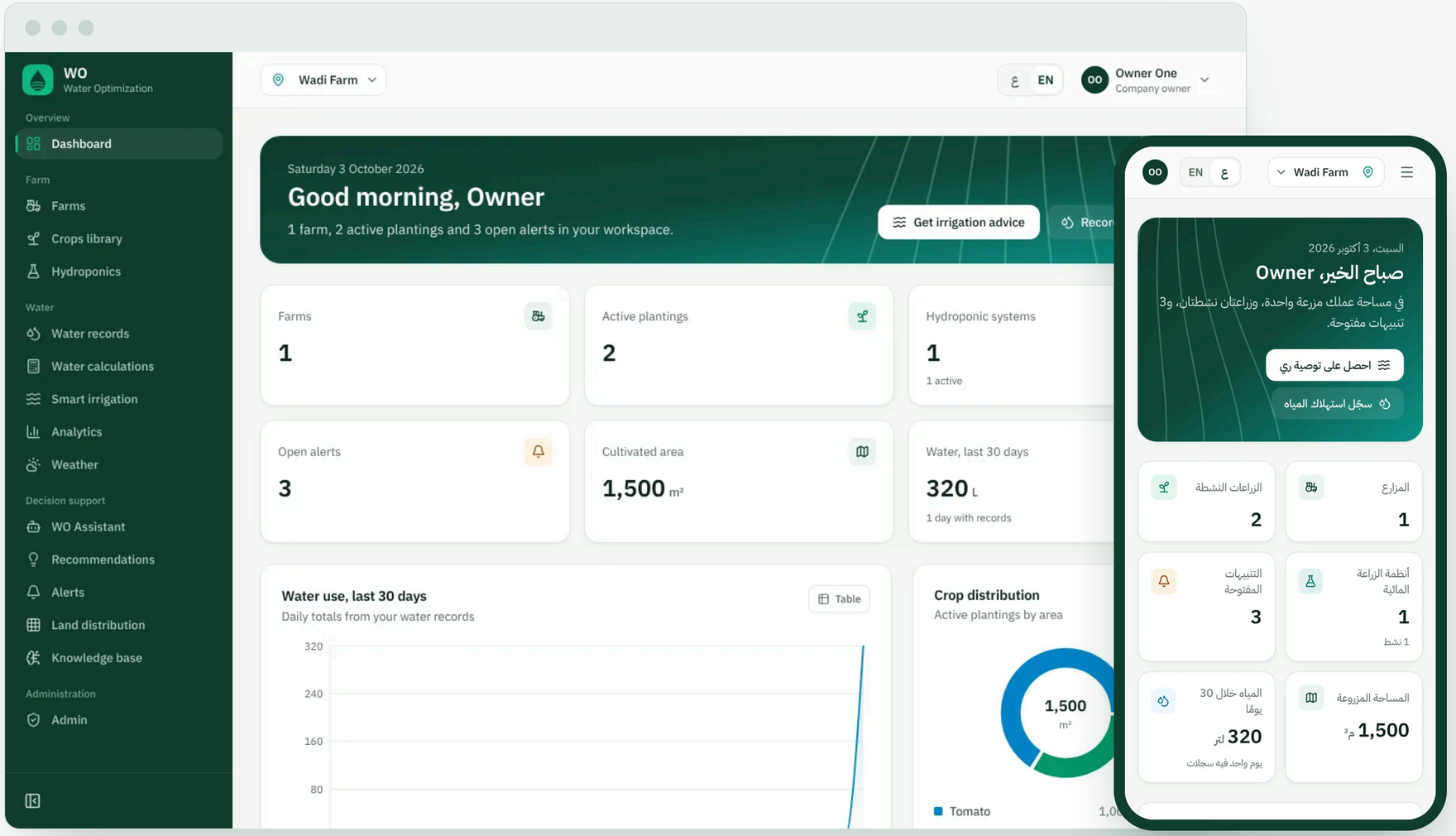
Water is the binding constraint: 4,491,285 of 4,500,000 L used; tomato is at its maximum area.

Algorithm	Production (kg)	Nodes expanded	Max frontier	Guarantee
A*	**32,000**	**7**	**29**	optimal
Uniform-cost	32,000	271	911	optimal
Greedy best-first	30,000	4	10	none
Breadth-first	6,000	55	217	none

Heuristic h = remaining units × p_max − fractional-knapsack upper bound (land- and water-relaxed). It never overestimates, so A* is optimal for the selected objective.

Example instance computed with WO's search code: 10,000 m² in 20 units, 4.5 million L for the season, FAO-56 seasonal water (ET0 5.5 mm/day, drip 0.90), illustrative yields entered as user inputs. Result statement: "Optimal under the selected objective and constraints."

Dashboards show real numbers, or an empty state



From the database

Counts, last-30-day water, crop areas, alerts, weather age.

Statistics need history

Trends appear after 7 days with records.

Arabic first

Full RTL on desktop and phone.

Measuring impact without overclaiming

> **WO does not claim to save water. Savings are computed only from a measured baseline that the farm records itself.**

Calculated

Requirement

`Σ FAO-56 gross L`

What the crop needed by FAO-56 on the recorded days.

Actual

Usage

`Σ recorded L`

What the water records say was applied.

Apparent

Savings

`calculated − actual`

May reflect under-irrigation or unrecorded use. Not proof of savings.

Measured

Savings

`baseline − current (L/m²/day)`

Only with a recorded baseline and at least 7 days of records. Otherwise: "Not available — record a baseline."

Implemented in water_service.impact (GET /api/water/impact). Weather and crop-stage differences between periods are not normalised, and the report says so.

Security lives in the backend

Permission	Individual	Company owner	Farm manager	Employee	Super admin
Read farm data	✓	✓	✓	✓	✓
Add records, run AI	✓	✓	✓	✓	✓
Edit farms and plantings	✓	✓	✓	—	✓
Delete records	✓	✓	✓	—	✓
Delete farms	✓	✓	—	—	✓
Assign members, manage users	—	✓	—	—	✓
Edit the knowledge base	—	—	—	—	✓

From services/access.py and the role checks in the services.

Another company's farm: 404

Not 403, so existence is never leaked.

JWT with revocation

HS256, 2 h expiry; logout bumps `token_version`.

Argon2id passwords

Login limited to 10 attempts per 5 minutes.

Uniform errors, no stack traces

Security headers, CSP, CORS from environment.

36 audited actions

Logins, records, recommendations, KB edits; secrets never logged.

What WO shows

Course concepts, working

- Knowledge-based agent, TELL/ASK (ch. 7)
- Forward and backward chaining, DPLL, TT-ENTAILS
- BFS, UCS, Greedy, A* with an admissible heuristic (ch. 3)
- CSP with AC-3, MRV, LCV, forward checking (ch. 6)

Built to be trusted

- Every recommendation explains itself
- Every coefficient cites its source
- Missing data is named, never guessed
- No savings claimed without a measured baseline

Next steps

- Root-zone soil water balance (FAO-56 ch. 8)
- Optional sensor input for soil moisture and flow
- More sourced hydroponic profiles
- Weather-normalised impact evaluation

Thank you